

Assignment-1: Shakespeare → LM → WordNet → Embeddings

Group: _____ Tiny-Shakespeare (GitHub) | 200 sents | 3272
toks | V=339

1. Tokenize → IDs (→ token_ids.csv)

```
toks = word_tokenize(sent.lower())      # NLTK, keep A-Z only
tok.build_vocab(tokenized)              # min_freq=2, <pad>=0, <unk>=1
ids = tok.encode(toks)                  # e.g. [hear,me,speak] -> [43,1,93]
```

2. Bigram language model (→ language_model.json)

```
P(w | prev) = (C(prev, w) + 1) / (C(prev) + V)  # add-1, V = 339
PPL = exp(-mean(log P))                        # train PPL = 107.55
```

3. WordNet similarity, keep > 0.5 (→ wordnet_similarity.csv)

```
from nltk.corpus import wordnet as wn
from itertools import combinations          # top-80 content words, no stopwords
s = wn.synsets(a)[0].wup_similarity(wn.synsets(b)[0]) # path ~0 pairs!
kept = [(a, b, s) for a, b in combinations(words, 2) if s and s > 0.5]
```

110 pairs > 0.5. Top:

w1	w2	wup
done	made	1.000
man	sir	0.947
city	rome	0.900
like	make	0.900
people	poor	0.889

4. Top-5 embedding (→ top5_embeddings.csv)

```
# window-2 co-occurrence on same corpus, L2-norm -> 50-d (seed 42)
mat[i, j] += 1 for neighbours in window      # 5 pairs -> 10 words
cos(a, b) = a.b / (||a|| * ||b||)            # e.g. sir-like 0.837
```

5. Iterations. 1 baseline (whitespace, $k=0$, path: 1 pair) → 2 hygiene (NLTK+lower, $k=1$: 0 pairs) → 3 **path**→**wup** (0→110) → 4 random→co-occurrence → 5 freeze + plot. Run: **make run** (venv only).